

Tyto-PhishShield

Demo 1

FiveGuys | 2026/05/22

Turn Your Team Into a
Human Firewall

Our team



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Problem

AI phishing scams are on the rise
and employees need to be trained to
detect these scams

Solution

Tyto-PhishShield is an
enterprise-grade Human Risk
Management platform

Project plan

A vertical approach with a strong base

01

Documentation

02

Frontend design

03

API gateway

04

Vertical slicing

Sprint Planning

- Agile Framework (Scrum, 2-week Sprint cycles)
- Sprint 1 Focus (Sprint planning, backlog commitments, and structural component development)
- Sprint 2 Focus (Feature integration, peer-reviewed pull requests, and client validation)
- Progress Tracking: Centralised Kanban board mapping issues

Sprint Planning

COS301-SE-2026 / Projects / Tyto-PhishShield Project Tasks

Backlog

Priority board

Roadmap

Team items

My items

New view

Filter by keyword or by field

Q Type [] to search

+ ▾

Insights

Workflows 7

⋮

Add status update

View

Backlog

1 / 10

Estimate: 0

This item hasn't been started

Tyto-PhishShield #97
(backend)report endpoint

Ready

0 / 5

Estimate: 0

This is ready to be picked up

In progress

3 / 10

Estimate: 22

This is actively being worked on

Tyto-PhishShield #26

UC-1: Users can report a suspicious email.

P0 5 M

Tyto-PhishShield #28

UC-10: System can send a scheduled simulated phishing campaign email.

P1 9 L

0 / 2

0%

Tyto-PhishShield #27

UC-6: Admin can control and schedule campaigns.

P1 8 L

In review

3 / 5

Estimate: 3

This item is in review

Tyto-PhishShield #31

UC-13: User receives XP update after an action

P2 3 S

Tyto-PhishShield #50

Add mailing-service With Basic Functionality

Tyto-PhishShield #51

API Gateway To Send Test Email

Done

56

Estimate: 40

This has been completed

Tyto-PhishShield #3

Complete use case diagrams

P0 1 L

Tyto-PhishShield #38

Complete Figma design

P0 5 M

Tyto-PhishShield #8

Create Design Specifications draft

P0 3 M

Tyto-PhishShield #6

Create test cases for each user story

P0 M

Tyto-PhishShield #21

Set up scrum week 1 document

P0 S

Tyto-PhishShield #13

Research Office.js API capabilities & Outlook

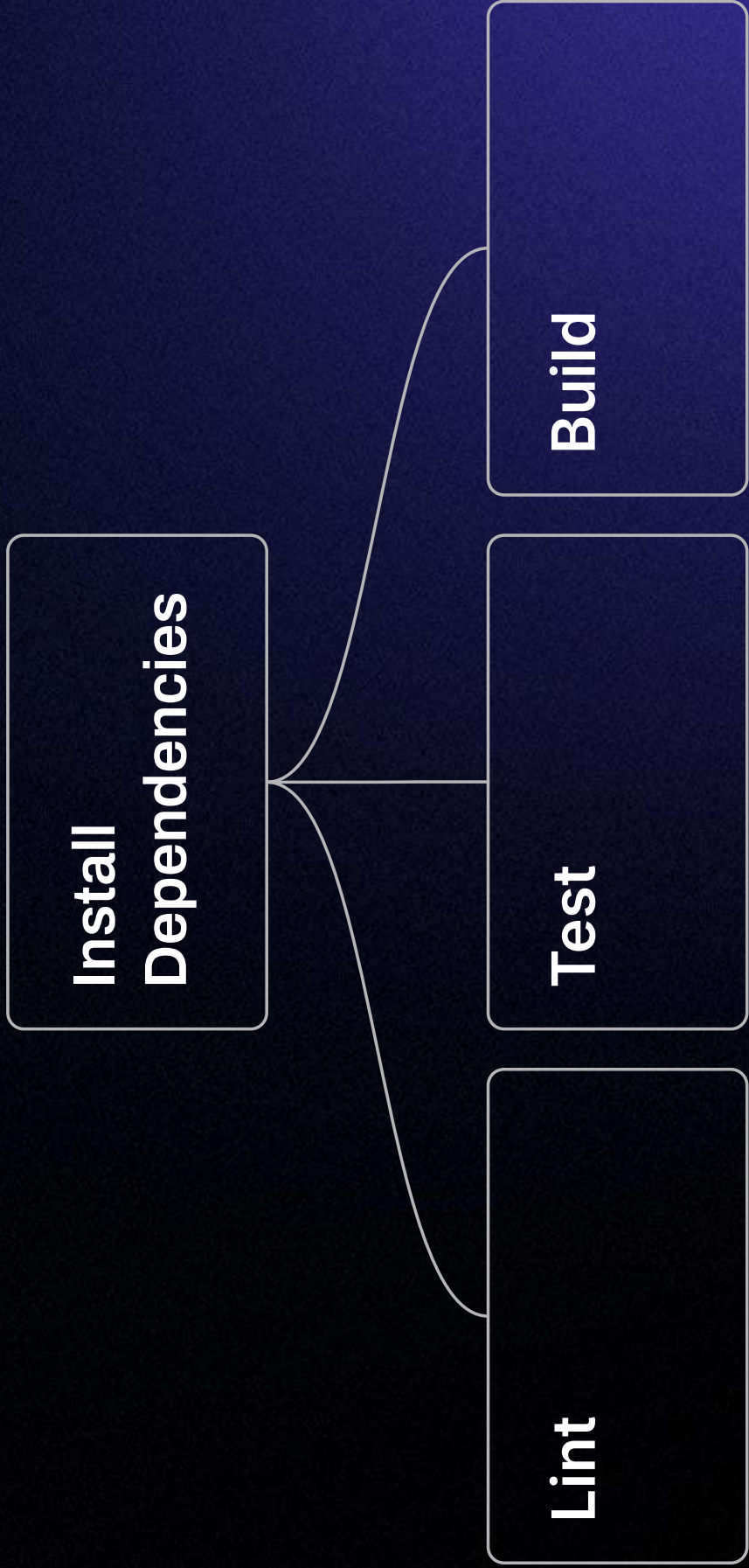
Key Use Cases

Primary Use Cases:

- * Administrators launch adaptive phishing campaigns
- * System automatically awards user reporting
- * Users can register and login to the system

Demo Time!

Continuous Integration



Triggered via pull request 15 hours ago

 Josua-Louw opened #95 dev

Status

Success

Total duration

1m 20s

Artifacts

1

ci-backend.yml

on: pull_request

Metric: Install dependencies

1 job completed

Show all jobs

Metric: Accounts Service Backend...

1 job completed

Show all jobs

Metric: Accounts Service Backend...

1 job completed

Show all jobs

Metric: Accounts Service Backend...

1 job completed

Show all jobs

Metric: API Gateway Backend Int...

1 job completed

Show all jobs

Metric: API Gateway Backend Int...

1 job completed

Show all jobs

Metric: API Gateway Backend Int...

1 job completed

Show all jobs

Metric: Billing Service Backend ...

1 job completed

Show all jobs

Metric: Billing Service Backend ...

1 job completed

Show all jobs

Metric: Billing Service Backend ...

1 job completed

Show all jobs

Metric: Billing Service Backend ...

1 job completed

Show all jobs

Testing Architecture & Stack



Sandwich Testing

- Top Down + Bottom Up (Hybrid)
- Demonstrates Core Value Immediately



Testing Scope

- Unit + Integration Testing (Backend)
- Unit Testing (Frontend)
- Validates Communication



Tooling Stack

- Vitest (React Frontend)
- Supertest (Backend)
- Jest (Outlook-addin + Backend)

Execution & Reliability

Why Passing Tests Matter

- Primary Safety Net
- Smoother Integration Process
- Smoother Deployment



FRONTEND
TESTS

100%
25/25

BACKEND
TESTS

100%
93/93

Code Coverage Analysis

BACKEND
Coverage

73%

FRONTEND
Coverage

64%

Why Coverage Matters

- Objective Progress Metric
- Smoother Integration Process

Repository Setup

Branching strategy: GitFlow model separating development from production releases

Workflow: Feature isolation *via* feature/ branches, peer-reviewed before merging into frontend/backend → develop



Monorepo Folder Structure

- Unified monorepo isolating docs, frontend, and backend
- Independent microservices as distinct containerised root directories.

```
Tyto-PhishShield
├── .gitignore
├── README.md
├── Docs/ ─── [ System Documentation ]
│   ├── Research/ ─── (Architecture, Auth0, POPIA/GDPR, OfficeJS)
│   ├── Scrum/ ─── (Project History, Sprints, Retrospectives)
│   └── SRS/ ─── (Specs, User Stories, Test Cases)
├── Project/ ─── [ Functional Codebase ]
│   ├── Backend/
│   │   ├── API_Gateway/ ─── (Port 3001: Edge Router & Auth0 RBAC Guard)
│   │   └── Accounts_Service/ ─── (Port 3002: Core Business Logic & DB Layer)
│   └── Frontend/
│       ├── Outlook_Add-in/ ─── (Native OfficeJS Add-in Component)
│       └── Website/ ─── (React 19 Administrative Dashboard)
```


ANY QUESTION?



Thank you!